B.TECH DATA SCIENCE

Module 4: Machine Learning Fundamentals

B.Tech Data Science — Expanded Study Notes and Exam Preparation

Contents
Four chapters of expanded concepts, definitions, examples, applications, methods and revision questions.
Includes side headings on every study page and important questions at the end for exam preparation.

Prepared for learning and exam revision. Follow your university syllabus and faculty guidance where they differ.

1.1 Introduction to Machine Learning

Study focus: Machine Learning Fundamentals. Read the side headings, understand the example, and revise the key ideas.

Definition

Machine learning (ML) is a field in which algorithms learn patterns from data to make predictions, classifications or decisions. It is useful when a fixed set of hand-written rules is inadequate or difficult to maintain.

Features and target

Features are the input variables used by a model. A target or label is the outcome to be predicted in supervised learning. The target must match the intended real-world decision.

Training and prediction

During training, an algorithm adjusts its internal parameters using examples. During prediction, the fitted model receives new inputs and produces an output.

Generalisation

A useful model performs well on relevant unseen cases, not only on its training examples. Generalisation is assessed using data held out from model fitting.

Applications

ML applications include spam filtering, demand forecasting, image recognition, recommendation, fraud detection and equipment maintenance. Suitability depends on data quality, cost, risks and the value of the decision.

1.2 Types of Machine Learning

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Supervised learning

Supervised learning uses examples with known target labels. Classification predicts categories, while regression predicts numerical values.

Unsupervised learning

Unsupervised learning looks for structure without a provided target label. Clustering groups similar observations, while dimensionality reduction creates a smaller representation of data.

Reinforcement learning

Reinforcement learning studies an agent choosing actions and receiving rewards through interaction with an environment. The objective is to learn a policy that maximises long-term reward.

Semi-supervised and self-supervised learning

Semi-supervised approaches combine a small labelled set with a larger unlabelled set. Self-supervised methods create learning targets from the data itself, often to build useful representations.

Choosing a type

The learning setup depends on the available labels, goal, feedback process and evaluation method. A task should not be called supervised if no suitable target is defined.

1.3 ML Workflow

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Problem framing

Define the prediction target, intended use, prediction time, unit of analysis and success metric. Clarify what action will follow the model output.

Data preparation

Inspect missing values, inconsistent labels, duplicate records and sampling limitations. Separate features from the target and ensure that inputs would be available in real use.

Baseline

Build a simple baseline such as predicting the majority class or a typical numerical value. A complex model is only useful if it improves meaningfully on an appropriate baseline.

Training and tuning

Fit the model on training data and select settings using validation data or cross-validation. Avoid using the final test set repeatedly to choose the model.

Evaluation and deployment

Evaluate the selected model on untouched test data, review error types and limitations, then plan deployment and monitoring if the model will be used operationally.

1.4 Regression and Classification

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Regression

Regression predicts a numerical quantity such as house price, temperature or delivery duration. Errors can be measured in the target's units or using squared-error measures.

Classification

Classification predicts a category such as spam/not spam or a risk group. Models may output class labels, probabilities or scores that require a decision threshold.

Decision threshold

A classifier's threshold changes the balance between false positives and false negatives. The threshold should reflect the costs of mistakes and operational constraints.

Examples

Predicting tomorrow's electricity demand is regression; deciding whether a message is spam is classification. The same domain can support both kinds of task.

Task framing matters

A numerical code for categories does not automatically make the task regression. The meaning of the target determines the task and evaluation measure.

2.1 Linear Regression

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Purpose

Linear regression models a numerical outcome using a linear combination of input features. In the one-feature case, the form is y-hat = b0 + b1x.

Intercept and slope

The intercept b0 is the predicted value when x is zero, when that interpretation is meaningful. The slope b1 describes the change in predicted outcome for a one-unit increase in x under the fitted model.

Fitting

Ordinary least squares selects coefficients to minimise the sum of squared residuals on the training data. A residual is the observed value minus the predicted value.

Assumptions and limitations

Interpretation and statistical inference may rely on assumptions about the relationship, errors and sampling. Outliers, nonlinear patterns, multicollinearity and leakage can affect results.

Example

A model may predict electricity use from temperature and time of day. Good practice includes plotting residuals, comparing against a baseline and evaluating on a later or separate dataset.

2.2 Logistic Regression and Classification

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Purpose

Logistic regression is commonly used for classification, especially binary classification. It models a probability through a logistic function rather than directly fitting an unrestricted numerical output.

Probability output

The predicted probability lies between 0 and 1. A threshold converts the probability into a class, but that threshold should be selected with the task's error costs in mind.

Interpretation

Coefficients relate features to the model's log-odds under the model specification. Their meaning depends on feature units and whether other features are held constant.

Multiclass problems

Extensions can handle more than two classes. Class imbalance and differences in error costs should be considered when selecting a model and threshold.

Evaluation

Use a confusion matrix, precision, recall, F1-score and other relevant measures. Accuracy alone can be misleading when one class is much more common.

2.3 Decision Trees and Random Forests

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Decision tree

A decision tree splits observations using feature-based rules. Classification trees predict categories; regression trees predict numerical values. Trees can be visualised and are often easier to explain than complex models.

Splitting and depth

The algorithm chooses splits that improve a criterion. Very deep trees can fit noise, so depth, minimum leaf size or pruning may be used to control complexity.

Random forest

A random forest combines many trees trained with randomness in samples and feature selection. Aggregation often improves stability compared with one deep tree.

Advantages and limitations

Trees can model nonlinear interactions and require relatively little scaling. Large forests are less transparent, may be computationally heavier and still require careful validation.

Practical use

Compare tree-based models against a baseline and simpler alternatives. Inspect errors, feature behaviour and performance on relevant subgroups before deployment.

2.4 K-Nearest Neighbours, Naive Bayes and SVM

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K-nearest neighbours

KNN predicts from the labels or values of nearby training observations. The choice of k, distance measure and feature scaling can strongly affect performance.

Naive Bayes

Naive Bayes applies Bayes' rule with a conditional-independence assumption among features given the class. Despite the simplifying assumption, it can work well for some text and classification tasks.

Support vector machine

An SVM seeks a boundary that separates classes with a large margin; kernel methods can represent nonlinear boundaries. Feature scaling and parameter selection are important.

Comparison

KNN can be simple but expensive at prediction time; Naive Bayes is efficient but uses a strong assumption; SVMs can be powerful but may need careful tuning.

Selection

No algorithm is best for every dataset. Compare candidate models with the same split and metric, and consider interpretability, cost and operational requirements.

2.5 Clustering and K-Means

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Clustering

Clustering groups observations based on a similarity or distance definition without using target labels. The resulting groups require interpretation and are not automatically meaningful categories.

K-means idea

K-means assigns observations to k clusters and updates each cluster centre to reduce within-cluster squared distances. It alternates between assignment and centre-update steps.

Choosing k

The number of clusters must be chosen. An elbow plot or silhouette score can provide evidence, but domain usefulness and stability also matter.

Scaling and outliers

Distance-based clustering is sensitive to feature scale and extreme values. Standardisation or robust preprocessing may be appropriate when justified.

Limitations

K-means favours roughly compact clusters and requires numerical features and a selected k. Different initialisations can produce different solutions, so repeated runs may be needed.

3.1 Training, Validation and Test Sets

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Training set

The training set is used to fit model parameters. Any learned preprocessing steps, such as scaling or imputation, must be fitted using training data only.

Validation set

Validation data helps choose model settings, features and thresholds. Repeatedly tuning against it can lead to over-specialisation to the validation set.

Test set

The test set is reserved for final evaluation after decisions are made. Repeatedly using it to tune models weakens its role as an independent estimate.

Cross-validation

Cross-validation rotates which part of the data is held out for validation. It can use data efficiently, but the split design must match the task and data structure.

Time and grouped data

For forecasting, use time-aware splits that respect chronology. If many rows belong to the same person or device, grouped splitting may be needed to avoid leakage across sets.

3.2 Overfitting, Underfitting and Regularisation

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Overfitting

Overfitting occurs when a model captures noise or training-specific patterns. It often performs very well on training data but poorly on unseen data.

Underfitting

Underfitting occurs when a model is too simple or poorly trained to capture useful patterns. It may perform poorly on both training and test data.

Bias-variance trade-off

A model's error can reflect systematic simplicity, sensitivity to the training sample and unavoidable noise. Increasing complexity may reduce one problem while increasing another.

Regularisation

Regularisation penalises overly complex parameter values or structures. Other remedies include reducing model complexity, improving data quality, using cross-validation and obtaining more representative data.

Learning curves

Comparing training and validation performance as data size or complexity changes can help diagnose fit problems. Diagnosis should consider the task and the amount of available data.

3.3 Feature Engineering and Data Leakage

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Feature selection

Feature selection chooses inputs relevant to the task. Useful features can improve efficiency and interpretation, but selection must not use information from the final test outcomes.

Scaling and encoding

Algorithms such as KNN and SVM are sensitive to scale. Categorical variables may require encoding, while tree-based models have different preprocessing needs.

Leakage examples

Leakage can occur when a feature is recorded after the outcome, when target-derived statistics include validation records, or when duplicate cases appear in both training and test sets.

Pipeline approach

Use a reproducible pipeline that learns imputation, scaling and encoding from training data and applies the same transformations to future records.

Review

Ask what would be known at the moment of prediction, how the data was collected and whether the evaluation resembles real deployment.

4.1 Regression Metrics

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Mean absolute error

MAE = average of |actual – predicted|. It expresses average absolute error in the target's units and is relatively easy to explain.

Mean squared error

MSE averages squared errors. Squaring penalises large errors more strongly and expresses the result in squared target units.

Root mean squared error

RMSE is the square root of MSE and returns to the target's units. It is sensitive to large errors.

Metric choice

MAE may be useful when a typical absolute error matters; RMSE may be useful when large mistakes deserve stronger penalty. Compare metrics on the same target scale and test data.

Residual analysis

Plot or summarise residuals to find systematic patterns, changing spread and unusual cases. A single metric can hide important failure modes.

4.2 Classification Metrics and Confusion Matrix

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Confusion matrix

A binary confusion matrix contains true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN). Define the positive class explicitly.

Accuracy

Accuracy = (TP+TN)/(TP+TN+FP+FN). It measures the share of correct predictions but can be misleading when classes are imbalanced.

Precision and recall

Precision = TP/(TP+FP), the share of predicted positives that are correct. Recall = TP/(TP+FN), the share of actual positives detected.

F1-score

F1 is the harmonic mean of precision and recall: 2×precision×recall/(precision+recall), when the denominator is nonzero. It balances the two but does not reflect every cost.

ROC-AUC and threshold

ROC-AUC measures ranking across thresholds. A metric does not determine the operational threshold by itself; false-positive and false-negative costs must be considered.

4.3 Responsible and Explainable ML

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Fairness

Evaluate whether model errors or outcomes differ across relevant groups. Differences can arise from representation, measurement, historical processes or model behaviour.

Privacy and security

Use appropriate access controls, minimise sensitive information and protect data and model outputs. Models can sometimes reveal information about training data, so privacy risk deserves attention.

Explainability

Explainability methods help describe why a model produced an output. Explanations are approximations in many cases and should not be treated as proof of causation.

Human oversight

High-impact uses may need human review, appeal routes and clear accountability. A model should not replace professional judgement where its limitations are significant.

Documentation

Record intended use, data scope, metrics, limitations, subgroup results, version and monitoring plan. Responsible use continues after deployment.

4.4 Deployment and Monitoring

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Deployment forms

Models may be used in batch jobs, dashboards, APIs or embedded systems. The form depends on latency, reliability, integration and maintenance requirements.

Data drift

Data drift occurs when input data patterns change. Concept drift refers to a change in the relationship between inputs and outcomes. Either can reduce performance.

Monitoring

Monitor input distributions, prediction patterns, latency, failures and outcome metrics when labels become available. A technically running model may still be producing poor predictions.

Retraining

Retraining should be based on evidence, quality checks and a defined process rather than an automatic response to every small fluctuation. Validate the new version before replacing the old one.

Rollback and governance

Maintain versioning, logging, ownership and a rollback plan. Decide who can approve changes and how incidents will be investigated.

5. Important Exam Questions and Revision

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Long-answer questions

1. Differentiate supervised, unsupervised and reinforcement learning. 2. Compare regression and classification. 3. Explain the ML workflow and data splitting. 4. Explain overfitting and underfitting with remedies. 5. Explain confusion matrix, precision, recall and F1-score. 6. Discuss responsible deployment and monitoring.

Short-answer questions

1. Define feature and target. 2. What is a baseline? 3. Define cross-validation. 4. What is regularisation? 5. Define data leakage. 6. State two clustering applications. 7. What is data drift?

Numerical/practical questions

1. Calculate MAE and RMSE for a set of predictions. 2. Calculate accuracy, precision, recall and F1 from a confusion matrix. 3. Choose an appropriate metric for imbalanced classification. 4. Describe a time-aware split for forecasting.

Revision checklist

Review learning types, regression, classification, common algorithms, training/validation/test data, overfitting, metrics, fairness, privacy, deployment and monitoring.

Exam tip

For algorithm questions, write the objective, working principle, example, advantages and limitations. For metric problems, define TP, TN, FP and FN before substituting into formulas.